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Lab Number: 2

ELEN 50L

5 October 2023

Th 2:15-5PM

Part 1:

1. Resistance Value Table

	R_1	R_2	R_3	R_4	R_5	R_6	
Nominal Values	1.0	3.9	8.2	12	15	20	kΩ
Measured Values	.990	3.838	8.205	11.962	14.869	19.955	kΩ

	I1 (mA)	V1 (V)	V2 (V)
Therotical (using nominal)	2.0	2.0408	7.9592
Theoretical (using measured)	2.1	2.0505	7.9495
Measured by DMM	2.070	2.052	7.947
Percent Error	-1.43%	.07%	-.03%

6. V1/V2=0.2582, R1/R2=0.2579

They are very close

8.

	I1 (mA)	I2 (mA)	I3 (mA)	V1(V)	V2(V)	V3(V)
Theo (nom R)	2.7	1.9	.8848	2.745	7.255	7.255
Theo (Meas R)	2.8	1.9	.8841	2.746	7.254	7.254
Meas from DMM	2.772	1.889	.884	2.747	7.252	7.252
Percent Error	1%	.58%	.01%	-.04%	.03%	.03%

9. Parallel of R2 and R3 = 2.615 K-Ohms. The added parallel resistor actually dropped resistance and made current increase but voltage drop.

10. I2/I3=2.1369, R2/R3=2.1378; These values are related by being really close (the same in theory)

12.

	I1 (mA)	I2 (mA)	I3 (mA)	I4 (mA)	V1 (V)	V2 (V)	V3 (V)	V4 (V)
Theo (nom R)	3.1586	1.7542	0.8343	0.5701	3.1586	6.8414	6.8414	6.8414
Theo (meas R)	3.2	1.8	0.8340	0.5721	3.1571	6.8429	6.8429	6.8429
DMM	3.186	1.783	.834	.572	3.158	6.841	6.841	6.841
PE	-.44%	-.94%	0%	-.02%	.03%	-.03%	-.03%	-.03%

Part 2:

	I1 (mA)	I2 (mA)	I3 (mA)	I4 (mA)	I5 (mA)	Vs (V)	V1 (V)	V2 (V)	V3 (V)	V4 (V)	V5 (V)	V6 (V)
Theo (Nom R)	2.3941	1.9502	.4439	.3305	.1133	10	2.3941	7.6059	3.6396	3.9663	1.6699	2.2665
Theo (meas R)	2.4	2.0	.4421	.3306	.1136	10	2.4001	7.5999	3.6448	3.9551	1.6887	2.2664
DMM	2.424	1.981	.444	.3306	.1138	10.0005	2.4017	7.5981	3.6421	3.9528	1.6865	2.2641

KCL a: -2.424 +1.981+.444=.001A

KCL b: +.444 - .3306+.1138= -.0004A

KCL c: -1.981-.444+2.424=.001A

KCL d: -.444+.3306-.1138=-.0004A

KVL 1: -10.0005+2.4017+7.5981=-.0007

KVL 2: -7.5981 +3.6421+3.9528=-.0032

KVL 3: -3.9528+1.6865+2.2641= -.0022

KVL 4: -10.0005 +2.4017+3.6421+3.9528=-.0039

KVL 5: -10.005 +2.4017+3.6421+1.6865+2.2641= -.0061

5. Req= 4.122K-Ohms vs theoretical which was 4.177

6. Calculated Req = 10.0005/(2.424)=4.126K-Ohms

7. We could measure V6 and I6 and find R6 through Ohms law (V/I=R).